

Learning science

The pedagogy in Continuum is code, not copy. Every function here lives in `packages/domain`, which imports no database client and calls no model, so each rule below is unit tested as arithmetic.

Mastery is four numbers, not one

A single "progress" percentage cannot distinguish a learner who has read a page from one who can apply the idea to a problem they have never seen. Continuum tracks four dimensions per concept in

`learning_states` :

Dimension	What raises it
exposure	Reading a lesson or completing a resource
understanding	Guided practice, with diminishing returns
transfer	Correct answers to unseen checkpoints
retention	Holding up over time, decaying between reviews

Plus `confidence`, tracked separately and never mixed into the others, because the gap between confidence and transfer is itself the useful signal.

continuum

Home

Ask

Plan

LEARN & BUILD

Study

Build

RESEARCH

Projects

Library

Context

Review

Settings

YOUR GOALS

Master SQL and Python... 58%

Build a leakage-conserv... 34%

Raise SAT score from 15... 42%

Complete and publish O... 71%

Jump to anything

Mukilan

CBSE Class 12

Continuum

Study

Search workspace

Ask

Saved

Study

5 concepts 3 practice sets

Due today

3 concepts are due

Ordered by what keeps slipping, not by what is oldest.

3 days

Parameterized queries (%)

Forgotten 2 times — it comes back sooner each time

Review

Cursor fetch semantics

Forgotten 1 time — it comes back sooner each time

Review

SAT advanced geometry — circles & parabolas

Forgotten 1 time — it comes back sooner each time

Review

Continue

Electric potential

You have studied this, but recall is slipping — transfer 28%.

transfer 28%

Review this

Concepts

Electric potential

transfer 28%

Never practised

Study

Cursor fetch semantics

transfer 60%

Practised yesterday

Study

Parameterized queries (%)

exposure 60%

Practised today

Study

commit() and rollback()

transfer 70%

Practised 2 days ago

Study

SAT advanced geometry — circles & parabolas

recall 67%

Practised today

Study

Material and practice

Practice sets

New set

From a photo

Material

Find material

Mastery per concept, with the evidence behind it rather than a time-based percentage.

Reading never raises transfer

```
if (evidence.kind === "lesson_read") {
  next.exposure = Math.max(next.exposure, 0.8);
  next.status = current.status === "misconception_detected" ? current.status : "exposed";
  next.explanation = "Lesson exposure was recorded; transfer did not change because no
independent evidence was provided.";
}
```

This is the rule most study tools get wrong. Time spent is not learning, and a progress bar that fills as you scroll is measuring scrolling. Here, `transfer` is untouched by exposure, and the state carries a written explanation of why.

Retention decays with time, not with use

$$r' = \text{clamp}(r - \min(0.18, 0.008 \cdot \Delta t_{\text{days}}))$$

```
const days = (reviewTime - previousReview) / 86_400_000;
next.retention = clamp(current.retention - Math.min(0.18, days * 0.008));
```

Roughly 0.8 percentage points per day, capped at 18 points per gap so a long absence does not zero a concept that was genuinely learned.

Unseen evidence, weighted by quality

When an unseen checkpoint is passed with $s \geq 0.7$:

$$q = s \cdot c \cdot (0.7 + 0.3d) \cdot h \cdot \text{dim}$$

$$t' = \text{clamp}(t + 0.08 + 0.24q), \quad r' = \text{clamp}(r + 0.05 + 0.13q)$$

where s is score, c completeness, d difficulty, h a hint penalty of 0.68 when a hint was used, and dim a diminishing-returns factor:

$$\text{dim} = \max(0.55, 1 - 0.035 \cdot \max(0, n - 2))$$

for n prior pieces of evidence. The tenth correct answer on the same concept moves the needle less than the third, which is what you want from a system trying to decide whether to move on.

Mastery is a conjunction, and it is strict

$$\text{mastered} \iff n \geq 4 \wedge t \geq 0.78 \wedge r \geq 0.68 \wedge u \geq 0.8$$

```
const repeatedEvidence = next.evidenceIds.length >= 4;
next.status = repeatedEvidence && next.transfer >= 0.78 && next.retention >= 0.68 &&
next.understanding >= 0.8
  ? "mastered" : "practicing";
```

Four separate pieces of evidence, high transfer, and retention that has survived a gap. One good day does not produce "mastered".

Failure names a misconception rather than lowering a score

```
} else {  
  next.transfer = clamp(next.transfer - (0.04 + 0.08 * (1 - score)));  
  next.status = "misconception_detected";  
  next.explanation = "The unseen checkpoint exposed a persistent misconception; targeted review  
was scheduled.";  
}
```

`misconception_detected` is sticky: later exposure and practice cannot silently clear it, because the status guard preserves it. It clears when the learner passes an unseen checkpoint on the thing they got wrong.

The screenshot displays the Continuum learning platform interface. On the left is a sidebar with navigation options: Home, Ask, Plan, Learn & Build (Study, Build), Research (Projects, Library, Context), Review, and Settings. Below these are 'Your Goals' with progress bars for 'Master SQL and Python...' (58%), 'Build a leakage-conserv...' (34%), 'Raise SAT score from 15...' (42%), and 'Complete and publish O...' (71%). The main area shows a goal 'Raise SAT score from 1520 to 1570+' with a progress bar at 42% and 65 days left. Below the goal is a 'Next milestone' box: 'Reach consistent 800 Math practice' (1 of 4 milestones complete) with an 'Open the plan' button. A 'CONCEPT MAP' section titled 'See the branches—not a manufactured chain' shows a grid of tasks. A task 'Review the last full-length mock (Bluebook #4)' is highlighted, showing its status (NOT STARTED), prerequisites (Foundations Final), and mastery/confidence levels (0%). A sidebar on the right for this task shows 'Not started', 'Mastery 0%', 'Confidence 0%', 'Attempts 0', 'Lesson Not completed', 'Last reviewed Not reviewed', and 'Response None yet'. It also lists 'Prerequisites' (Start here), 'Unlocks' (Path milestone), and a 'Recommended next action' (Open the lesson) with an 'Open lesson' button.

Concepts, prerequisites and what each one unlocks, with mastery and confidence shown separately.

Spaced repetition: SM-2 with two departures

The file opens by stating the gap it closed:

Continuum already measured mastery in four dimensions and then scheduled nothing with it, which left a student with a number and no answer to the only question that matters day to day: *what should I review now?*

Departure 1: recognition does not advance the interval

Standard SM-2 takes a self-reported grade from 0 to 5. Self-report is exactly the signal that inflates: a learner who has just re-read a page feels fluent and grades themselves 5.

Here the grade is derived from evidence:

```
export function gradeFrom(evidence: {
  correct: boolean;
  unseen: boolean;
  explanationScore?: number; // 0-1 from an explain-back attempt
  seconds?: number;
}): ReviewGrade {
  if (!evidence.correct) return "forgot";
  if (evidence.explanationScore !== undefined && evidence.explanationScore < 0.5) {
    // Right answer, cannot explain it. That is recognition, and it is exactly
    // the case a self-reported grade would call "easy".
    return "hard";
  }
  if (!evidence.unseen) return "hard";
  if (evidence.seconds !== undefined && evidence.seconds > 90) return "good";
  if (evidence.explanationScore !== undefined && evidence.explanationScore >= 0.85) return
  "easy";
  return "good";
}
```

A correct answer to a question already met is capped at `hard`. It moves the schedule forward because the memory is real, but it never earns the long interval that only transfer to a new item justifies. Correct but slow is `good`, not `easy`. Correct but unable to explain is `hard`.

Departure 2: a lapse costs ease but never resets the record

Standard SM-2 sends a lapsed card back to a one-day interval and discards everything learned about it. That is punishing and it throws away information.

$$I' = \max(1, \text{round}(I/2))$$

The interval halves and the ease drops, so a shaky concept returns soon without pretending it was never learned.

The schedule

$$e' = \text{clamp}(e + \delta_g, 1.3, 3.2), \quad \delta = \begin{cases} -0.24 & \text{forgot} \\ -0.14 & \text{hard} \\ 0 & \text{good} \\ +0.12 & \text{easy} \end{cases}$$
$$I' = \begin{cases} 1 & \text{reps} = 1 \\ 3 & \text{reps} = 2 \\ \text{clamp}(\text{round}(I \cdot e' \cdot m_g), 1, 180) & \text{otherwise} \end{cases} \quad m = \begin{cases} 0.6 & \text{hard} \\ 1 & \text{good} \\ 1.15 & \text{easy} \end{cases}$$

The 180-day ceiling has a stated reason: beyond that, a scheduled review is indistinguishable from "you know it".

Every interval carries a sentence

```
export interface ReviewOutcome extends ReviewState {  
  dueAt: string;  
  /** Plain language, for the UI. Never a number without a reason beside it. */  
  because: string;  
}
```

Which produces, from real state:

```
"You had this at 12 days. Back to 6 days."  
"Right, but not yet fluent – back in 4 days."
```

A scheduler that says "review this on Tuesday" and cannot say why is asking for trust it has not earned.

Explain-back

`explanationScore` in the grader comes from the explain-back surface: the learner writes the idea in their own words and the grading distinguishes a correct answer from a correct answer they can reconstruct. That distinction is what separates `hard` from `easy` above, and it is the difference between recognition and recall that self-report cannot see.

Grading is conservative by construction

The question-bank grader runs a deterministic comparison first and only calls a model when the answer is genuinely ambiguous. When it does, reconciliation is asymmetric:

```
const confirmed = deterministicCanConfirm(answer, expected) || (single ? single.verdict ===  
"correct" : false);  
const downgrade = deterministic.correct && !confirmed;
```

A single evaluator may lower a deterministic pass. It may never award one. This exists because the grader once marked a textbook misconception "Correct": a model was willing to be generous about parameter order, and a student would have walked away with the error confirmed. Being conservative about a right answer costs a learner one extra review. Being wrong about a misconception costs them the concept.